Andrew Phengthalasy

aphengthalasy@wpi.edu | (401) 428-6881 | *Linkedin* | *Github*

Education

Worcester Polytechnic Institute **Worcester, MA**
Bachelor of Science: Computer Science | **Minor(s):** Financial Technology & Data Science *May 2028*
Relevant Coursework: Data Modeling & Analysis, Financial Analytics, Software Engineering, Computational Methods, Database Systems I, Algorithms, Systems Programming Concepts, Object Oriented Design Concepts

Skills

Programming & Web: Java, Python, SQL, Typescript, R, C/C++, HTML/CSS, NodeJS, Javascript, React, Mantine, Tailwind
DevOps: Git, Jira, Agile/Scrum, Auth0, Supabase, PrismaORM, Docker, Nginx, Postman, Power Automate, Render, Excel
Data & AI: Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn, Statsmodels, Jupyter Notebook, Mistral AI

Projects

Content Management System | *Hanover Insurance | PERN, Typescript, Mantine*
- Built a **full-stack** document, user, workflow, and role management system, **automating 12+** key workflow controls with a **10 person team.**
- Refactored backend architecture & integrated **50+ API calls** into multiple employee management pages, tested via **Postman.**
- Supported a **Mistral AI-powered Chatbot,** capable of document & employee retrieval.
- Utilized **I8next**, leading internalization efforts & translated the UI into **10** languages for accessibility.

Sector Returns During Recessions Analysis | *Python, Pandas, FRED API, Yahoo Finance*
- Built an **E2E ETL Pipeline** to scrape, clean & align **324 months** of sector-level excess returns, **NBER** recession dummies, & macroeconomic **INDPRO, MKT** controls.
- Executed Newey-West **HAC OLS** Models (**12 lags**), via statsmodels producing market betas from **0.27 to 1.27** with strong statistical significance.
- Delivered model diagnostics confirming **recession coefficient p > 0.05 a**cross six sectors, rejecting the hypothesis.

Data Structures Suite | *Linked Lists, Deques, and Algorithmic Analysis*
- Developed a comprehensive **Java-based** data structures suite including a generic **doubly-linked list**, **deque**, and **sorting algorithm** performance analyzer.
- Created **50+** JUnit test cases to validate functionality, edge cases, and exception handling for all data structures.
- Performed empirical algorithm analysis to compare runtime performance of **15+ sorting algorithms.**
- Visualized run time results in an **excel spreadsheet** to confirm theoretical time complexities.

Housing Price Prediction Model | *Python, Scikit-learn, Pandas, Seaborn*
- Created a **data preprocessing pipeline** for 545+ **records,** converting categorical variables into quantitative data.
- Implemented **Simple Linear**, **Multiple Linear**, and **Lasso Regression**, resulting in an **89%** improvement in accuracy.
- Quantified model performance via $\mathbf{R^2}$ and **MSE** to conclude feasibility of price predictions.

Related Experience

Research Assistant | *Research Solutions Institute (WPI)* *April 2025 - Present*
- Maintained a centralized **Spin Based Opportunity database (infoEd)** to streamline management of **100+ faculty research** proposals and grant proposals.
- Automated administrative workflows by designing standardized templates, reducing proposal preparation time by **15%.**
- Collaborated with a cross-functional RSI team to identify and implement **software solutions** to improve research **efficiency**.

Fundraising Intern | *Society of Asian Scientists & Engineers (WPI)* *March 2026 - May 2026*
- Strategized **3+** meaningful fundraising efforts, raising **3 times** the annual amount of the previous year
- Partnered with **6 Executive Board** members to streamline event logistics for maximum efficiency.
- Analyzed **fundraising data** from the general body and public to evaluate **valuable insights** for continuous improvement in future efforts.

Involvements

Layout & Logistics Manager | *Pan Asian Association* *September 2025 - May 2026*
PASS-CS Scholar | *NSF & Computer Science Department* *August 2024 - May 2028*
Student Worker | *Housing & Residential Experience Center* *September 2024 - Present*